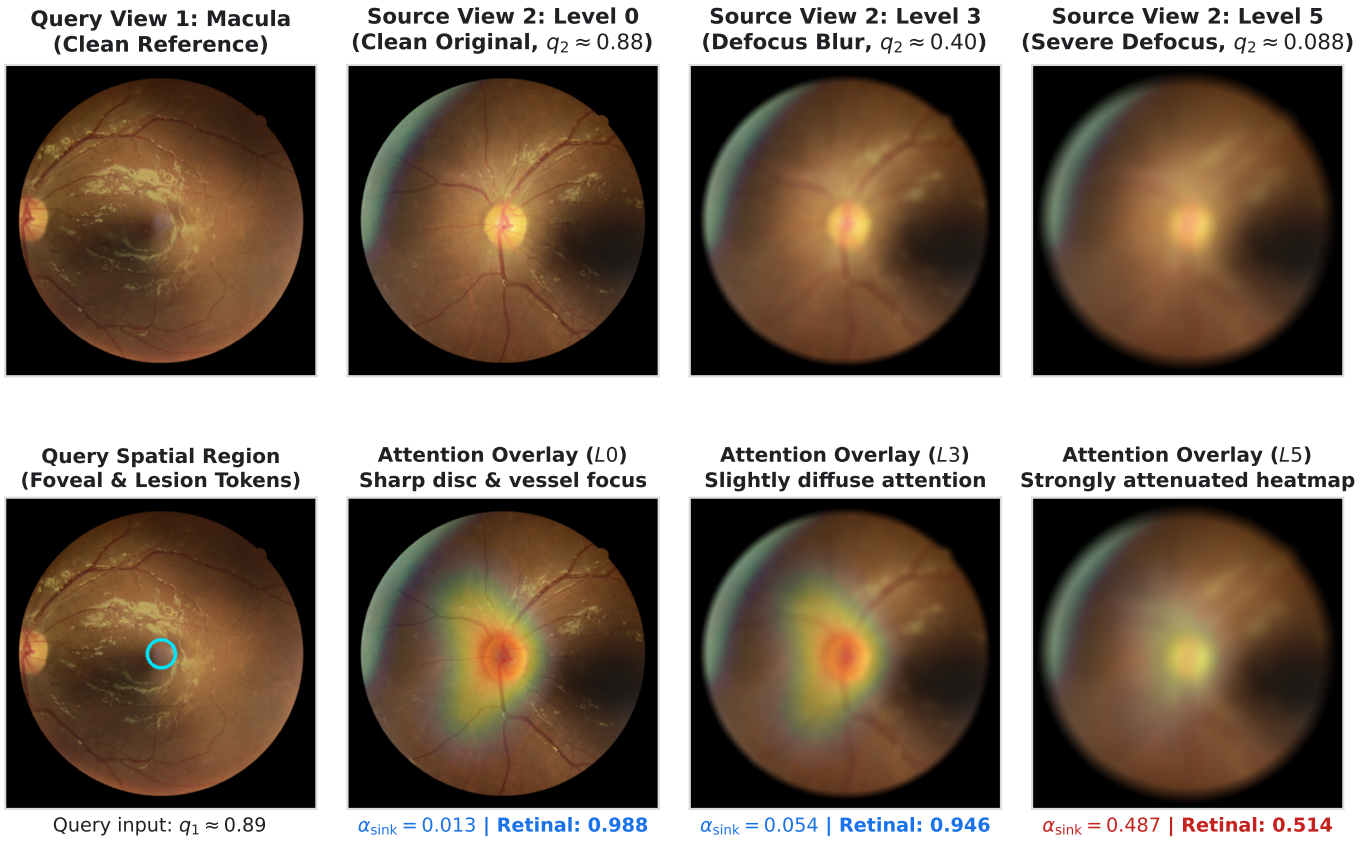
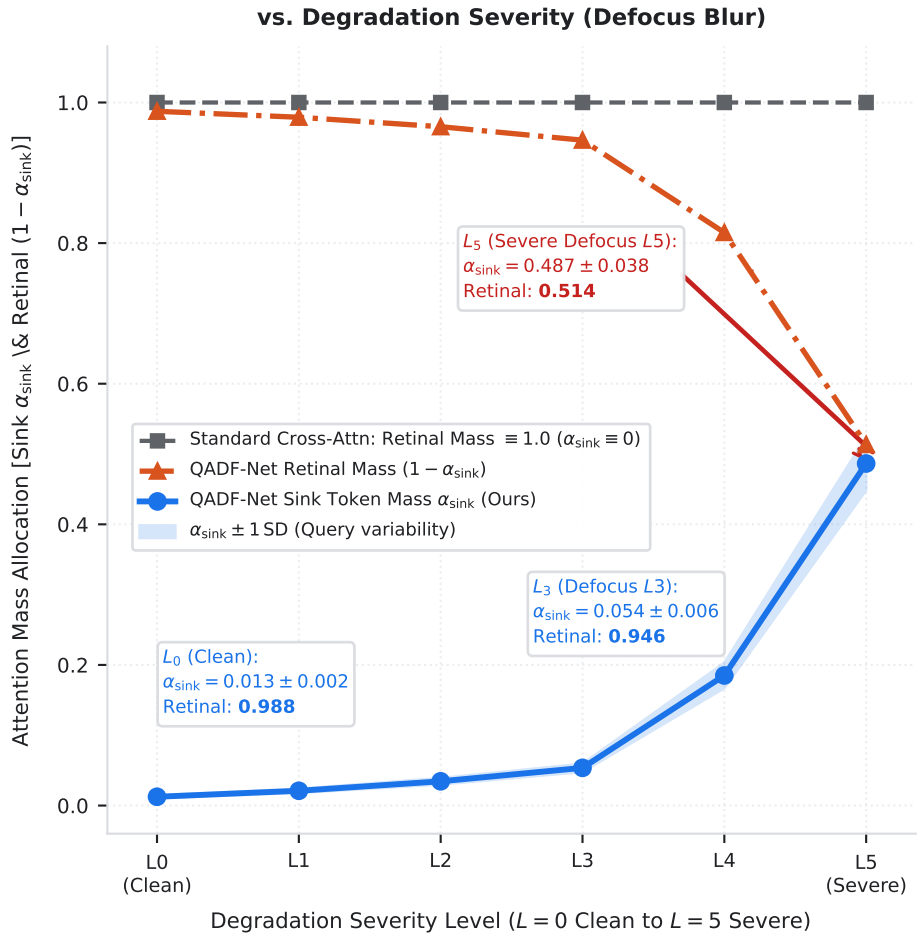

Query View 1: Macula
(Clean Reference)
Source View 2: Level 0
(Clean Original, $q_2 \approx 0.88$)
Source View 2: Level 3
(Defocus Blur, $q_2 \approx 0.40$)
Source View 2: Level 5
(Severe Defocus, $q_2 \approx 0.088$)
Query Spatial Region
(Foveal & Lesion Tokens)
Attention Overlay ($L0$)
Sharp disc & vessel focus
Attention Overlay ($L3$)
Slightly diffuse attention
Attention Overlay ($L5$)
Strongly attenuated heatmap
Query input: $q_1 \approx 0.89$
$\alpha_{sink} = 0.013$ | Retinal: 0.988
$\alpha_{sink} = 0.054$ | Retinal: 0.946
$\alpha_{sink} = 0.487$ | Retinal: 0.514


Cross-Attention Mass Redistribution
vs. Degradation Severity (Defocus Blur)
Attention Mass Allocation [Sink α_{sink} \& Retinal $(1 - \alpha_{sink})$]
1.0
0.8
0.6
0.4
0.2
0.0
Standard Cross-Attn: Retinal Mass ≡ 1.0 ($\alpha_{sink} \equiv 0$)
QADF-Net Retinal Mass ($1 - \alpha_{sink}$)
QADF-Net Sink Token Mass α_{sink} (Ours)
$\alpha_{sink} \pm 1$ SD (Query variability)
L_5 (Severe Defocus $L5$):
$\alpha_{sink} = 0.487 \pm 0.038$
Retinal: 0.514
L_3 (Defocus $L3$):
$\alpha_{sink} = 0.054 \pm 0.006$
Retinal: 0.946
L_0 (Clean):
$\alpha_{sink} = 0.013 \pm 0.002$
Retinal: 0.988
L0
(Clean)
L1
L2
L3
L4
L5
(Severe)
Degradation Severity Level ($L = 0$ Clean to $L = 5$ Severe)